Impact of Using PRS-CSx and Pruning and Thresholding for Polygenic Partitioning of Apparent Treatment Resistant Hypertension

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Background

- Apparent treatment-resistant hypertension (aTRH):
  - Failure to achieve blood pressure (BP) control on three or more antihypertensive medications, OR
  - Prescribed four or more medications regardless of control
- aTRH → greater risk for diabetes, chronic kidney disease, cardiovascular disease, and stroke
- Despite its clinical significance, the biological mechanisms underlying aTRH remain poorly understood
- Goal: k-means clustering of aTRH associated variants using two SNP selection approaches: PRS-CSx and Pruning & Thresholding (P&T)

Comparing two SNP selection strategies for clustering of aTRH associated variants to further understand the underlying mechanisms of treatment resistance.

Methods

Cohort	Cases	Controls
Million Veteran Program (MVP)	11,762 NHW	42,850 NHW
	5,071 NHB	11,091 NHB
	Total: 16,833	**Total: 53,931**
Vanderbilt University Medical Center's BioVU	3,541 NHW	39,356 NHW
	745 NHB	5,804 NHB
	Total: 4,286	**Total: 39,356**

Table 1. Number of MVP and BioVU cases and controls. Cases: failure to achieve controlled BP on 3 antihypertensives, one of which must be a thiazide diuretic, or prescribed 4 or more medications regardless of control. Controls: controlled HTN (<140/90 mmHg) on 1 or 2 medication classes. Excluding patients with chronic kidney disease (stages 4 and 5) and causes of secondary HTN.

1. aTRH Polygenic Score (PS) Development & P-Value Thresholding

PRS-CSx
1. Build weights: MVP aTRH sumstats
2. Build scores: BioVU genotype data
3. Optimal PS SNP p-value selection

Pruning & Thresholding
1. LD prune MVP aTRH sumstats
2. Build scores: BioVU genotype data
3. Optimal PS SNP p-value selection

Optimal P-value threshold: P < 1e-2

2. Defining aTRH Loci & Clustering Analysis

1. Filter MVP GWAS to SNPs with P < 1e-2
2. Subset 91 traits to same aTRH SNPs
3. Align 91 traits to aTRH risk allele
4. Calculate z-scores for the 91 traits
4. ClustImpute R package: impute missing data
5. NbClust R package: optimal number of non-overlapping clusters of SNPs

4 non-overlapping clusters of SNPs | **3 non-overlapping clusters of SNPs**

3. Cluster-Specific Pathway Analyses

1. Subset MVP aTRH MVP cross-population sumstats to cluster-specific SNPs
2. Utilize the SNP2GENE function of FUMA to perform gene mapping
3. Perform cluster-specific pathway and tissue enrichment analyses with the GENE2FUNC function of FUMA

Figure 1. Overview of methods. MVP GWAS refers to the cross-population meta-analysis of aTRH. BioVU data refers to the genotype data from BioVU (MEGA). PS: Polygenic scores. LD: Linkage disequilibrium.

Results

Identifying Biologically Meaningful Clusters of aTRH SNPs

- **PRS-CSx Clusters:**
  - Cluster 1: ⇧SBP/DBP, glucose, adiposity; ⇩cardiac and blood cells
  - Cluster 2: ⇧anthropometric, liver, lifestyle, BP, cardiac; ⇩lipids, women's health, blood cell traits
  - Cluster 3: weak associations overall, strong ⇩BP traits
  - Cluster 4: nearly inverse of Cluster 2
- **P&T Clusters:**
  - Cluster 1: broadly ⇧glycemic, adiposity, BP, liver, lifestyle
  - Cluster 2: broadly ⇩glycemic, adiposity, lipids, BP, blood cells
  - Cluster 3: ⇧BP, ⇩cardiac structure/function, adiposity, lipids, blood cell traits
- PRS-CSx: finer contrasts, P&T: broader, interpretable signatures

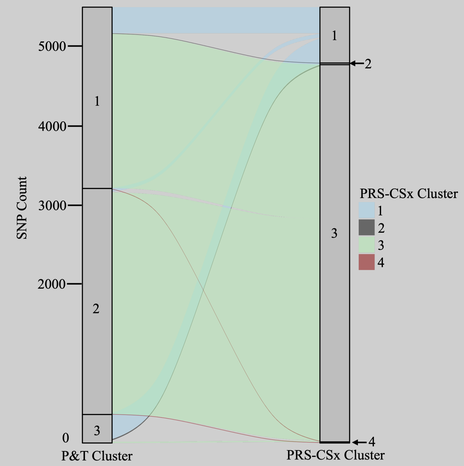

Figure 2. Alluvial plot showing the differences in distribution between P&T Clusters and PRS-CSx Clusters. The number of SNPs per cluster is shown next to each bracket. SNPs that were not shared between methods were excluded for plotting.

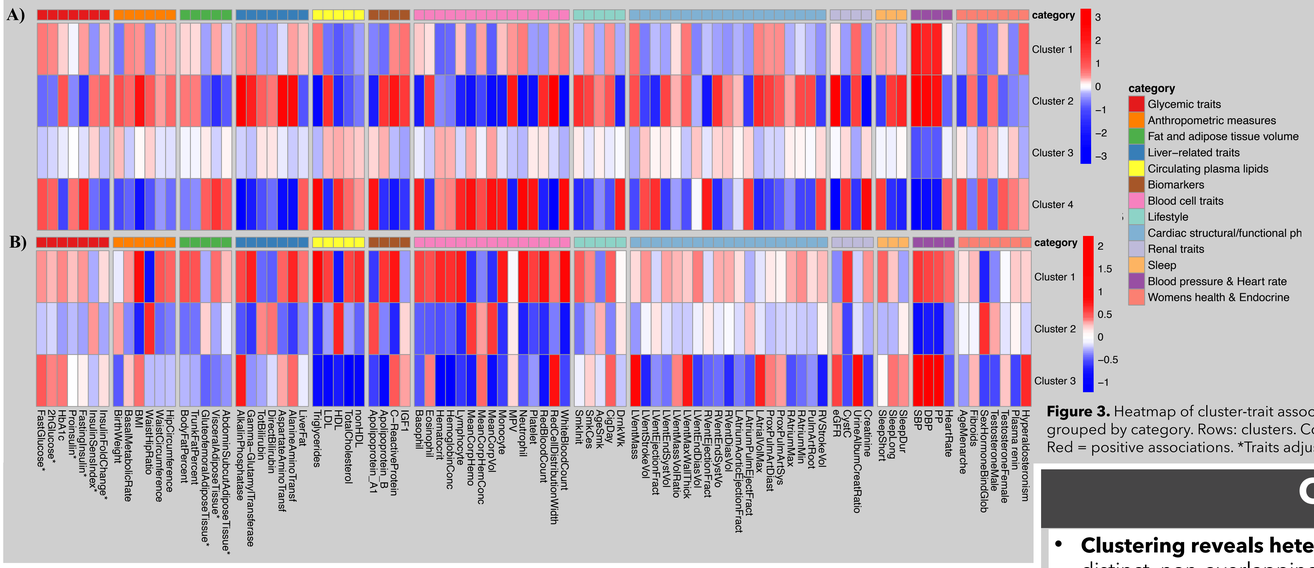

Figure 3. Heatmap of cluster-trait associations with aTRH P < 0.01. Columns: 91 cardiometabolic traits, grouped by category. Rows: clusters. Color intensity: direction and magnitude of association. Blue = negative, Red = positive associations. *Traits adjusted for BMI. **A)** PRS-CSx-based. **B)** P&T-based.

Cluster-Specific Pathway Analyses

Method	Cluster	Tissue	$P_{Bonferroni\text{-}adjusted}$
PRS-CSx	1	Bladder	5.15×10^{-7}
		Adrenal gland	7.82×10^{-7}
		Heart left ventricle	2.20×10^{-5}
	3	Bladder	5.90×10^{-5}
		Adrenal gland	1.10×10^{-3}
P&T	1	Whole blood	2.46×10^{-3}
	2	Bladder	4.98×10^{-8}
		Colon	2.92×10^{-7}
		Uterus	8.94×10^{-4}
		Prostate	4.66×10^{-2}
		Kidney	4.70×10^{-2}

Table 3. Cluster-specific FUMA GENE2FUNC results for tissue specificity. Only significantly ($P_{Bonferroni}$ < 0.05) enriched tissues are shown.

FUMA GENE2FUNC Tissue Specificity

Comparison: PRS-CSx highlighted a narrower set of tissues (endocrine and cardiac), whereas P&T captured a broader range of tissues and more significant associations, including immune, reproductive, and gastrointestinal systems.

FUMA GENE2FUNC Biological Pathways

Comparison: both methods highlight developmental pathways. PRS-CSx Cluster are enriched for hormonal/cardiac regulation, while P&T Clusters show overall stronger associations and are enriched for protein degradation and skeletal system-related pathways.

Method	Cluster	Gene Ontology Biological Pathway	$P_{FDR\text{-}adjusted}$
PRS-CSx	1	Anterior posterior pattern specification	7.33×10^{-3}
		Glucocorticoid biosynthesis process	1.87×10^{-2}
		Embryonic skeletal system development	1.87×10^{-2}
		Membrane repolarization during cardiac muscle cell action potential	3.61×10^{-2}
	3	Homophilic cell adhesion via plasma membrane adhesion molecules	1.32×10^{-3}
		Embryonic skeletal system development	3.08×10^{-2}
		Glucocorticoid biosynthetic process	4.33×10^{-2}
P&T	1	Ubiquitin dependent endoplasmic reticulum-associated degradation	2.91×10^{-2}
		Endoplasmic reticulum degradation	3.40×10^{-2}
	2	Embryonic skeletal system development	3.92×10^{-8}
		Embryonic skeletal system morphogenesis	2.41×10^{-7}
		Skeletal system development	2.41×10^{-7}
		Anterior posterior pattern specification	3.50×10^{-7}

Table 4. Cluster-specific FUMA GENE2FUNC results for enrichment of gene sets. Only gene sets with FDR adjusted P-value < 0.05 are shown. Top 4 of over 20 significant pathways are shown for PRS-CSx Cluster 1 and P&T Cluster 2.

Conclusions

- **Clustering reveals heterogeneity**: Both SNP selection strategies identified distinct, non-overlapping clusters with unique cardiometabolic association profiles, which supports multiple biological routes to treatment resistance
- **Clustering outcomes differ by method:**
  - PRS-CSx selected ~5x more SNPs, yielding 4 clusters
  - P&T: 3 clusters with more interpretable associations
- **Shared biological signals:** Both highlight adiposity/glycemic traits, developmental pathways, and hormonal regulation as contributors to aTRH
- **Method-specific biological signals:**
  - PRS-CSx: enriched for neurohormonal, developmental, and cardiac regulation pathways
  - P&T: broader tissue enrichment, pathways linked to protein quality control and skeletal development
- **Method assessment:** P&T: clearer interpretability and stronger tissue and pathway signals → may be the preferred SNP selection strategy
- **Clinical implications:** Distinct genetic clusters suggest mechanisms that may enable pathway-informed precision therapies

Genetic clustering, with consideration of SNP selection method, has potential to further elucidate the mechanistic heterogeneity of aTRH.

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